3D Printing Filament Settings

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Material	Nozzle °C	Bed °C	Enclosure	Part cooling	Dry at
PLA	180-220	50-60	Not needed	Yes (100%)	45C / 4-6h (opt.)
PETG	220-250	70-85	Not needed	Yes (30-50%)	65C / 4-6h
ABS	220-250	100-110	Required	No / low	65-80C / 2-4h
ASA	235-255	90-110	Required	No / low	70-80C / 4h
PC	260-310	100-120	Required	No	80C / 6-8h
PEEK	370-420	120-160	Required	No	120-150C / at least 3h
Nylon PA6	230-260	70-90	Required	Minimal	70-80C / 8-12h
Nylon PA12	240-270	70-90	Required	Minimal	70-80C / 8-12h
PA-CF	260-280	80-100	Required	Minimal	70-80C / 8-12h
PETG-CF	240-260	80-90	Not needed	Yes (30-50%)	65C / 4-6h
PCTG	230-250	70-85	Not needed	Yes (30-50%)	65C / 4-6h
CPE	240-260	75-90	Recommended	Yes (low)	65-70C / 4-6h
HIPS	220-250	100-115	Required	No / low	65C / 4h (opt.)
PLA Silk	195-225	50-60	Not needed	Yes (50-70%)	45C / 4-6h (opt.)
PLA Matte	195-220	50-60	Not needed	Yes (100%)	45C / 4-6h (opt.)
PLA Wood	190-220	50-60	Not needed	Yes	45C / 4-6h (opt.)
PLA Metal	190-225	50-60	Not needed	Yes	45C / 4-6h (opt.)

Typical manufacturer-published ranges for each material class. Not measurements we took. Always check your filament maker's own figures.

What Goes Wrong (and Why)

The problem you are most likely to hit with each material, and why it happens.

Material	What goes wrong
PLA	Parts soften around 60C, so a print left on a dashboard will sag. If the part needs to survive heat, move to PETG or ABS rather than trying to tune PLA.
PETG	Stringing between towers. Drop the nozzle 5C at a time and raise travel speed before you touch retraction distance.
ABS	Corners lift off the bed and layers split partway up. The cause is nearly always a draught, not the bed temperature. Enclose the printer and turn the part cooling fan off.
ASA	Same warping profile as ABS, plus a stronger smell. Ventilate the room and keep the chamber warm and still.
PC	Layers delaminate under load even though the print looked clean. That is moisture. Dry the spool properly and keep it dry during the print.
PEEK	The realistic problem is machine capability, not settings. If your hot end tops out at 300C and the bed at 110C, PEEK is not an option regardless of tuning.
Nylon PA6	Steam pops and a furry surface mean the filament is wet. Nylon absorbs moisture faster than you can print it, so drying once is not enough - it has to stay dry during the print.
Nylon PA12	Poor bed adhesion. Nylon does not like PEI. A garolite or PA-specific sheet, or a glue stick layer, fixes what temperature tuning will not.
PA-CF	Underextrusion that gets worse as the print goes on. The carbon fibre has widened the nozzle bore. Fit a hardened steel or ruby nozzle before the first print, not after.
PETG-CF	People expect carbon fibre to mean stronger in every sense. It means stiffer. Impact resistance goes down, not up.
PCTG	Availability, more than printing. Fewer brands carry it, so colour choice is narrow compared with PETG.
CPE	CPE is often confused with chlorinated polyethylene, an unrelated industrial rubber. In 3D printing CPE always means co-polyester.
HIPS	Buying it as a support material without checking you have a second extruder. On a single-nozzle printer HIPS is just a weak ABS-like filament.
PLA Silk	The sheen disappears with too much cooling. Turn the fan down, print a little hotter and slower than plain PLA.
PLA Matte	The matting filler is mildly abrasive. A brass nozzle survives it, but expect faster wear than plain PLA.
PLA Wood	Clogging. Wood particles bridge a small nozzle easily. Use 0.4mm or larger and avoid long retractions.
PLA Metal	People expect metal strength. The filler adds weight and finish, and reduces strength compared with plain PLA.

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